

Self-Supervised Contextual Representations of Chest X-Ray Images

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Abstract—Self-supervised learning (SSL) has gained traction in medical image analysis, enabling representation learning with limited labels. While contrastive SSL, using diverse augmentations, has become dominant, we argue that applying standard augmentations, originally designed for natural images, to medical images like chest X-rays is suboptimal. Chest X-rays possess unique structures and subtle abnormalities that differ from natural images, and preserving these during augmentation is critical for learning clinically meaningful representations. In this paper, we introduce a novel set of domain-specific contextual transformations tailored for chest X-rays, including anatomy-aware perturbations and context random masking, designed to preserve diagnostic semantics during SSL pre-training. This augmentation strategy is the first to explicitly align transformation design with radiological context, addressing a major gap in existing medical SSL approaches. Experiments on NIH, RSNA, and SIIM datasets show that our approach yields up to 5% improvement in downstream tasks under limited supervision, compared to standard augmentations.

Index Terms—Self-supervised learning (SSL), chest X-ray, domain-specific transformations, NIH, RSNA.

I. INTRODUCTION

AI MODELS require high-quality annotated data, which is expensive in medical images. Self-supervised learning (SSL) [1], [2], [3], particularly discriminative approaches [4], [5], [6], [7], addresses this by learning view-invariant representations with limited labels. Various SSL methods have been applied to medical images [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], showing promise in classification and segmentation tasks. However, medical imaging poses unique challenges for discriminative SSL, which relies on data transformations to learn representations, invariant to changes in class-defining features [23], [24], [25], [26], [27]. While augmentations like cropping, color distortion, flipping, and masking are critical for SSL success [28], [29], [30], [31], they may not transfer well to X-rays or CTs due to high inter-class similarity and the subtle abnormalities [32], [33]. For instance, Fig. 1 compares augmented views of a dog and a chest X-ray. Natural images like dogs allow for diverse transformations in spatial arrangement, orientation, texture, and color. However, chest X-rays have

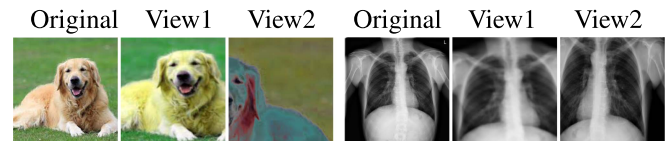


Fig. 1. Two augmented views are created via standard SSL transformations. Limited variability in X-rays, unlike natural images (e.g., dogs), poses unique challenges for SSL.

constrained spatial arrangements due to anatomical structures, uniform textures, and grayscale nature, limiting transformation variability. Medical images face additional challenges of data bias and subtle pathologies. Normal samples often outnumber abnormal ones, and transformations may further distort pathological features. This can lead SSL models to focus on general anatomical structures while overlooking important abnormalities during pre-training. While augmentation improvements have been proposed for natural images [30], [31], [34], similar advancements for medical images remain underexplored.

To address this, we propose domain-specific contextual transformations tailored for chest X-rays and the SSL framework based on the proposed contextual transformations. Within the domain of medical imaging, a range of SSL methods [8], [10], [11], [12], [13], [14], [15], [20] have been developed, all of which adhere to the fundamental principles of contrastive or non-contrastive SSL techniques. These methods share a common philosophy, augmented by additional components as evidenced in DiRA [11], CAID [20], Inter_Moco [15], and Preservational Contrastive Representation Learning (PCRL) [12], [13]. In line with the established paradigm, our proposed SSL framework adheres to these foundational principles and introduces a novel contribution in the form of domain-specific contextual transformations and their utilization within the proposed SSL framework. The aim is to emphasize the pathological conditions and enhance the diversity and informativeness in the augmented views. The proposed transformations highlight pathological lesions, such as tumors or nodules, introduce pathological noise patterns that mimic the characteristics commonly observed in chest X-rays, and leverage region-of-interest annotations, such as the lungs, to guide the transformations. By incorporating domain-specific data augmentations into the SSL pipeline, we can facilitate the learning of meaningful representations that capture both the general anatomical structures and the subtle pathological features. We performed multiple experiments on different chest X-ray datasets to validate the effectiveness and observed significant performance gains during downstream evaluations.

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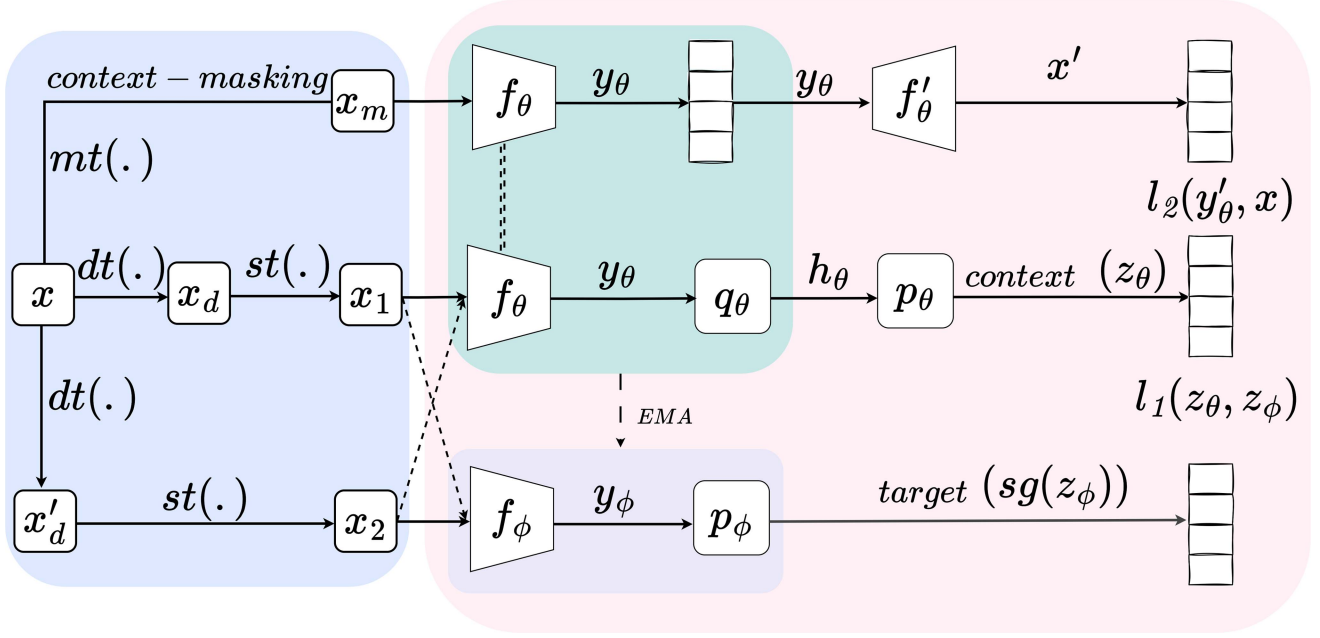


Fig. 2. Architecture of the proposed framework. $dt(\cdot)$, $st(\cdot)$, and $mt(\cdot)$ denote domain-specific, spatial, and masking transformations. Two augmented views x_1 and x_2 are generated by applying $dt(\cdot)$ and $st(\cdot)$ to x , while x_m is a context-masked view from $mt(\cdot)$. f_θ and f_ϕ are the context and target encoders, and f'_θ is the decoder.

II. METHODOLOGY

Fig. 2 illustrates our SSL framework comprising a context encoder (f_θ), target encoder (f_ϕ), and auxiliary decoder (f'_θ), along with domain-specific augmentations $dt(\cdot)$ and lung-focused context masking $mt(\cdot)$ tailored for chest X-rays.

Framework Overview: Given an input chest X-ray image x , we apply domain-specific transformations $dt(\cdot)$ to produce x_d and x'_d , followed by spatial transformations $st(\cdot)$ (e.g., random crop, affine distortions), resulting in two augmented views: x_1 and x_2 . These views are fed into the context encoder f_θ and target encoder f_ϕ , respectively. The encoders output feature representations y_θ and y_ϕ , which are then projected through multi-layer perceptron (MLP) heads q_θ and p_ϕ to produce h_θ and z_ϕ . Following practices in Bootstrap Your Own Latent (BYOL) [6] and SimSiam [35], an additional MLP p_θ further maps h_θ to z_θ for prediction alignment.

To enhance anatomical awareness, we introduce a third input view x_m via context random masking $mt(\cdot)$, applied within the lung fields of x . The masked input is passed through f_θ and decoded by f'_θ to reconstruct the original image as x' . This decoding path encourages spatial sensitivity in the representation by focusing reconstruction loss on the lung regions.

Domain-specific Contextual Transformations: As shown in Fig. 2, we first apply domain-specific transformations $dt(\cdot)$ to image x , generating intermediate views x_d and x'_d that simulate clinically relevant patterns such as opacities or anatomical noise. Standard spatial transformations $st(\cdot)$ (e.g., random cropping, flipping, affine distortions) are then applied to x_d and x'_d to obtain final views x_1 and x_2 , passed to the context and target encoders, respectively. This sequence, applying $dt(\cdot)$ before $st(\cdot)$, preserves the spatial integrity of simulated abnormalities, ensuring they remain localized within the lung regions. Reversing the order could distort or displace these patterns, reducing their clinical relevance.

The $dt(\cdot)$ module incorporates motion blur, anatomical noise, simulated pathologies, synthetic nodules, and opacity regions,

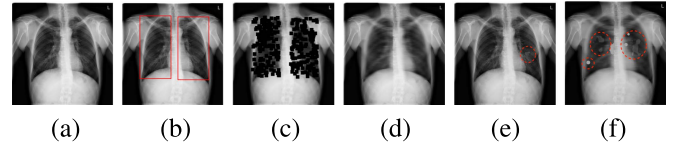


Fig. 3. Proposed domain-specific transformations. (a) Original X-ray (b) Identified lung regions (c) Context random masking (d) Motion Blur (e) and (f) having anatomical noise and simulated pathologies.

each applied within the left and right lung fields as defined by coarse boundary detection using HAAR (Haar-like Features), HOG (Histogram of Oriented Gradients), and LBP (Local Binary Patterns) filters, as illustrated in Fig. 3. Specifically, *motion blur* simulates respiratory or patient movement artifacts; *anatomical noise* adds random patterns resembling acquisition-related distortions; *simulated pathologies* introduce irregular patches mimicking masses or tumors; *nodule-like patterns* emulate small, localized abnormalities; and *opacity regions* replicate patchy consolidations seen in pneumonia or atelectasis. These variations expose the model to diverse, clinically relevant conditions during training while preserving anatomical context. The result is a pair (x_1, x_2) of semantically aligned but visually diverse views, essential for effective SSL.

We also introduce *context random masking* $mt(\cdot)$, which selectively masks random patches within the lung regions of the chest X-ray image, while leaving the surrounding anatomy intact. This operation encourages the model to focus on learning meaningful representations of lung structures by forcing it to infer missing contextual information from the unmasked regions. The approach aligns with radiological practice, where radiologists often concentrate on specific lung regions while also considering global anatomical context for interpretation. By simulating partial occlusion of clinically important regions, context random masking promotes robust, context-aware feature learning during pre-training.

Algorithm 1: Self-supervised Contextual Representation Learning.

Require: Chest X-ray image x ; transforms $dt(\cdot)$, $st(\cdot)$, $mt(\cdot)$; encoders f_θ , f_ϕ ; decoder f'_θ ; projection heads q_θ , p_θ , p_ϕ ; lung region masks P_r , P_l

- 1: $x_d, x'_d \leftarrow dt(x)$
- 2: Apply spatial transforms: $x_1 \leftarrow st(x_d)$, $x_2 \leftarrow st(x'_d)$
- 3: Generate masked view: $x_m \leftarrow mt(x)$
- 4: Compute embeddings: $y_\theta \leftarrow f_\theta(x_1)$, $y_\phi \leftarrow f_\phi(x_2)$
- 5: Compute projections: $z_\theta \leftarrow p_\theta(q_\theta(y_\theta))$, $z_\phi \leftarrow p_\phi(y_\phi)$
- 6: Reconstruct masked image: $\hat{x} \leftarrow f'_\theta(f_\theta(x_m))$
- 7: Compute similarity loss: $\mathcal{L}_1(z_\theta, sg(z_\phi))$
- 8: Compute reconstruction loss: $\mathcal{L}_2(\hat{x}, x)$
- 9: Compute total loss: $\mathcal{L} = \beta \mathcal{L}_1 + (1 - \beta) \mathcal{L}_2$
- 10: Update parameters of f_θ , q_θ , p_θ , f'_θ using $\mathcal{L}$
- 11: Update f_ϕ , p_ϕ via EMA of f_θ , p_θ

Loss Function: We update the parameters of f_θ using the cosine similarity loss defined in Eq. 1, which aligns the l_2 -normalized predicted embeddings z_θ and target embeddings $sg(z_\phi)$, where $sg(\cdot)$ is the stop-gradient operator:

$$\mathcal{L}_1(z_\theta, sg(z_\phi)) = \frac{z_\theta \cdot sg(z_\phi)}{\|z_\theta\| \|sg(z_\phi)\|} \quad (1)$$

The target encoder parameters ϕ are updated via exponential moving average (EMA) of θ . To promote spatially aware reconstruction, we introduce a weighted Mean Squared Error (MSE) loss (Eq. 2) that emphasizes lung-region pixels, improving fidelity in clinically important areas:

$$\mathcal{L}_2(\hat{x}, x) = \frac{1}{HW} \sum_{h,w} [\alpha (x_{h,w} - \hat{x}_{h,w})^2 \cdot \mathbf{1}_{(h,w) \in P_r \cup P_l} + (x_{h,w} - \hat{x}_{h,w})^2 \cdot \mathbf{1}_{(h,w) \notin P_r \cup P_l}] \quad (2)$$

Here, H and W are the image dimensions, and P_r , P_l represent pixels in the right and left lung regions. The final loss combines both terms as $\beta \mathcal{L}_1 + (1 - \beta) \mathcal{L}_2$, where $\alpha > 1$ encourages accurate reconstruction in lung areas. Algorithm 1 provides a step-by-step overview of our pre-training procedure.

III. EXPERIMENTAL SETUP

Datasets: For pre-training, we use Chest X-ray dataset [36], with 86,524 training and 25,596 test samples. For downstream evaluation, we perform: (i) **multi-label classification** of 14 thoracic conditions on NIH; (ii) **binary classification** (normal vs. pneumonia) on RSNA [37]; and (iii) **segmentation** of pneumothorax regions on SIIM-ACR [38].

Implementation Details: We pre-train with ResNet18 as backbone, 64 batch size, 100 epochs, and Layer-wise Adaptive Rate Scaling (LARS) [39] optimization, with 0.002 initial learning rate and $1e-5$ weight decay. Projection and prediction heads are three-layer MLPs (output: 256, hidden: 2048). Loss parameters: $\alpha = 2$, $\beta = 0.6$. Downstream evaluation uses a linear classifier (NIH, RSNA) and a decoder (SIIM-ACR segmentation). We assess performance via linear probing (LP) (frozen backbone) and semi-supervised fine-tuning (FT).

Baselines: We compare against supervised approaches with random (Sup.1) and pre-trained (Sup.2) initialization; general SSL methods including SimCLRv2, MoCov3, BYOL, and VICReg; and chest X-ray-specific SSL methods such

TABLE I
DOWNSTREAM EVALUATION ON NIH AND RSNA UNDER LP; WE REPORT AUC FOR NIH AND ACCURACY FOR RSNA TEST SETS

Methods	NIH				RSNA			
	1%	10%	30%	100%	1%	10%	30%	100%
Sup.1	51.6	57.1	61.1	61.8	75.6	76.8	77.0	78.2
Sup.2	59.6	66.1	68.1	69.8	77.1	79.4	80.3	81.1
SimCLRv2 [42]	62.7	66.7	68.5	69.5	76.8	80.0	81.1	81.4
MoCov3 [43]	62.8	70.0	72.1	73.1	76.8	80.1	80.8	81.2
BYOL [6]	63.3	70.6	72.8	73.5	76.5	78.4	80.1	80.9
VicReg [7]	61.1	65.7	68.2	69.3	76.7	80.4	80.9	81.2
PCRLv1 [12]	59.6	65.0	70.5	72.2	76.3	79.5	80.2	81.1
DiRA [11]	60.8	68.6	72.6	74.1	76.9	81.2	81.5	82.0
CAiD _{MoCo} [44]	60.7	67.9	70.3	73.5	79.0	81.0	81.4	82.0
PCRLv2 [13]	61.9	68.6	72.6	73.8	76.8	80.3	80.5	81.1
Inter_MoCo [15]	52.4	65.9	68.9	70.3	76.9	81.2	81.5	82.0
MLVICX [40]	60.1	66.1	69.1	71.0	78.3	81.2	81.9	82.0
CoBoom [41]	65.1	72.0	73.6	74.8	80.0	81.7	82.1	82.7
Ours	66.5	73.4	74.4	74.8	80.6	82.4	83.0	83.1

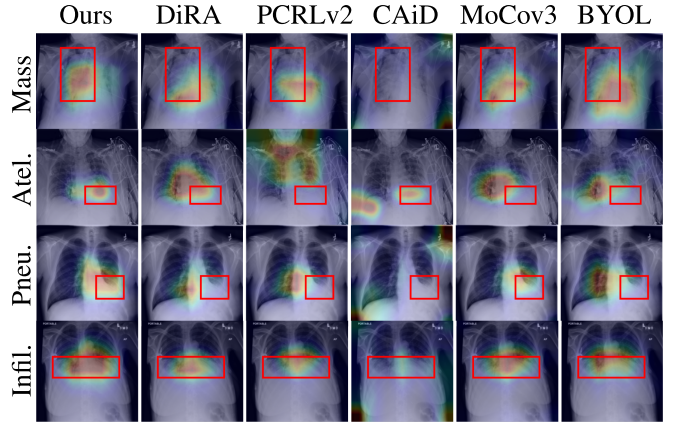


Fig. 4. Comparative heatmaps generated from the NIH dataset corresponding to the model trained with 1% labeled data showing Mass, Atelectasis, Pneumonia, and Infiltrate.

as PCRLv1 [12], PCRLv2 [13], DiRA [11], CAiD [20], MLVICX [40], and CoBoom [41].

IV. RESULTS AND DISCUSSION

Downstream classification (LP): Table I presents LP-based classification results on NIH and RSNA across varying label fractions. Our domain-specific transformations yield clear advantages, especially in low-data settings. With 1% NIH labels, we achieve over 3% AUC improvement over the best baseline; with 10% labels, the margin exceeds 2.5%. This trend continues at 30% and 100% label levels. On the RSNA dataset, our method also consistently outperforms all baselines in pneumonia classification. These results validate the effectiveness of our approach in label-scarce regimes.

Qualitative Results: Fig 4 shows heatmap visualizations from the NIH, using the model evaluated with 1% labeled data. Ground truth annotations, provided as bounding boxes, serve as a reference. Our approach demonstrates more accurate localization of pathological regions compared to baseline methods. Precise identification of conditions like Mass, Atelectasis, and Infiltrate has significant clinical implications for effective diagnosis. It highlights our method's effectiveness in correctly identifying and analyzing key ROIs in chest X-rays.

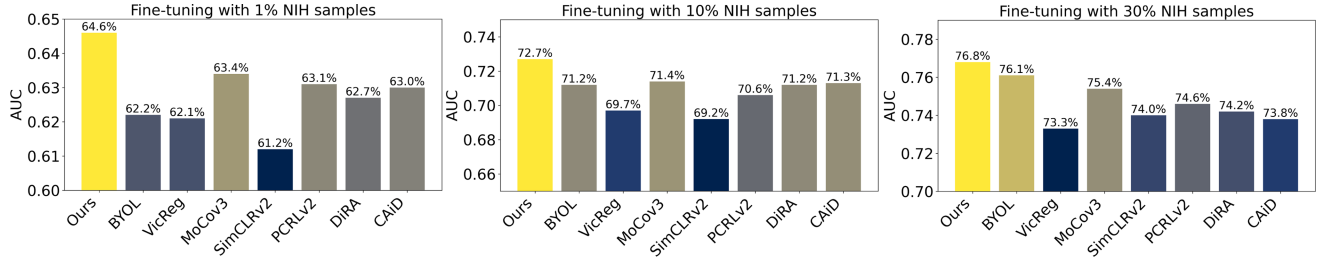


Fig. 5. Test set results in terms of AUC for downstream classification under the fine-tuning evaluations on the NIH dataset.

TABLE II
PNEUMOTHORAX SEGMENTATION UNDER LP AND FT EVALUATION PROTOCOLS
IN TERMS OF THE DICE SCORE AFTER THREE RUNS

Methods	SIIM-ACR Pneumothorax (Dice)			
	Linear Probing (LP)		Fine-tune (FT)	
	30%	All	30%	All
Sup.1	42.1 ± 0.1	48.4 ± 0.1	45.1 ± 0.4	51.2 ± 0.1
SimCLRv2 [4]	47.3 ± 0.3	52.3 ± 0.2	49.4 ± 0.3	54.1 ± 0.3
MoCov3 [43]	48.4 ± 0.3	53.1 ± 0.2	49.2 ± 0.1	54.1 ± 0.3
BYOL [6]	46.3 ± 0.2	50.7 ± 0.2	47.2 ± 0.4	52.3 ± 0.1
VicReg [7]	46.1 ± 0.2	48.4 ± 0.2	46.4 ± 0.3	54.2 ± 0.4
PCRLv1 [12]	43.3 ± 0.3	48.7 ± 0.1	46.4 ± 0.2	52.1 ± 0.2
CAiD _{MoCo} [20]	44.1 ± 0.3	55.3 ± 0.2	46.2 ± 0.1	56.4 ± 0.2
DiRA [11]	46.3 ± 0.3	55.5 ± 0.4	49.2 ± 0.1	57.3 ± 0.1
PCRLv2 [13]	46.4 ± 0.2	56.3 ± 0.1	46.8 ± 0.4	57.2 ± 0.2
Ours	53.2 ± 0.3	57.4 ± 0.26	52.4 ± 0.1	58.2 ± 0.2

Downstream classification (FT): Fig. 5 shows NIH dataset results when fine-tuning with varying amounts of labeled data. Our framework consistently outperforms baselines, especially under low-label settings. With only 1% labeled data, we achieve 64.6% AUC, while PCRLv2, CAiD, and MoCov3 reach around 63%, and BYOL, VICReg, and SimCLRv2 perform lower. At 10% labeled data, baselines average 70.5% AUC, whereas our method reaches 72.7%. With 30% data, all methods improve, with BYOL and our approach performing comparably, ours slightly ahead. These AUC scores highlight the effectiveness of our domain-specific transformations in learning discriminative representations.

Segmentation Results: Table II presents the findings of pneumothorax segmentation on the SIIM-ACR dataset corresponding to LP and semi-supervised FT using different subsets of labeled data. Note that using limited labeled data (30%), our approach outperforms the best baseline method, DiRA, with a considerable margin of 5% under LP. A similar superior performance is observed under semi-supervised FT with limited labeled data. The consistent superior performance signifies the importance of the proposed contextual transformation, which enforces the model to learn subtle localized representations along with generalized global features.

Discussion and Limitations: Our strong performance arises from domain-specific contextual transformations that enhance view diversity and guide representation learning toward lung anatomy. As shown in Table I and Fig. 5, we achieve 66.5% AUC with just 1% labeled NIH data, improving to 73.4% with 10% supervision. With 30% labels, fine-tuning yields further gains. This is driven by augmentations that inject anatomical noise and pathology-like patterns during pretraining, helping the

TABLE III
RESULTS OF AN ABLATION STUDY CONDUCTED WITH CONTEXT RANDOM
MASKING VIEW (x_m) AND THE DECODER MODULE (f'_θ) ON NIH DATASET
UNDER LINEAR PROBING EVALUATION

Method	x_m	f'_θ	α	NIH	
				1%	10%
ours	✓	✓	2	66.5	73.4
	✓	✓	1	64.7	72.6
	✗	✓	-	64.2	72.5
	✗	✗	-	63.1	71.8

model learn clinically relevant features. While effective, our augmentations are specific to chest X-rays and may not generalize to other modalities (e.g., CT or MRI). The approach also relies on coarse lung detection, which may be sensitive to occlusions or artifacts. In terms of complexity, the method adds minimal overhead to standard SSL frameworks, as transformations are lightweight and the decoder operates on sparse masked inputs. Future work will address generalizability and robustness across modalities.

V. ABLATION STUDIES

Table III shows an ablation study on two key components: context random masking (x_m) and the decoder module (f'_θ), using 1% and 10% labeled NIH data. All settings use the proposed domain-specific transformations for x_1 and x_2 . The full model achieves the best performance (66.5% AUC with 1%, 73.4% with 10%). Removing x_m lowers AUC to 64.2% (1%) and 72.5% (10%), while excluding both x_m and f'_θ further reduces performance (63.1% and 71.8%). This underscores the role of x_m and f'_θ in capturing contextual features and enhancing diagnostic accuracy. Additionally, setting the pixel-weighting factor $\alpha = 1$ in the reconstruction loss leads to a 1% AUC drop, validating the importance of emphasizing lung-region pixels during reconstruction.

VI. CONCLUSION AND FUTURE SCOPE

This paper introduces domain-specific contextual transformations for chest X-rays, focusing on lung anatomy. We propose augmentations such as motion blur, random noise, pathological patterns, and contextual masking aligned with diagnostic practices to foster self-supervised representation learning. Evaluations on multiple datasets for classification and segmentation using LP and FT protocols show notable gains over existing SSL methods. Future work will explore broader transformations to simulate diverse imaging scenarios and pathologies.

REFERENCES

- [1] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," *CoRR*, vol. abs/1807.03748, 2018. [Online]. Available: <http://dblp.uni-trier.de/db/journals/corr/corr1807.html#abs-1807-03748>
- [2] Y. Li, R. Wu, Q. Tan, Z. Yang, and H. Huang, "Masked spectral bands modeling with shifted windows: An excellent self-supervised learner for classification of medical hyperspectral images," *IEEE Signal Process. Lett.*, vol. 30, pp. 543–547, 2023.
- [3] J. Lin, Q. Cai, and M. Lin, "Multi-label classification of fundus images with graph convolutional network and self-supervised learning," *IEEE Signal Process. Lett.*, vol. 28, pp. 454–458, 2021.
- [4] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. 37th Int. Conf. Mach. Learn.*, 2020, vol. 119, pp. 1597–1607.
- [5] X. Chen, H. Fan, R. Girshick, and K. He, "Improved baselines with momentum contrastive learning," 2020, *arXiv:2003.04297*.
- [6] J.-B. Grill et al., "Bootstrap your own latent: A new approach to self-supervised learning," in *Proc. 34th Int. Conf. Neural Inf. Process. Syst.*, Dec. 2020, vol. 33, pp. 21271–21284.
- [7] A. Bardes, J. Ponce, and Y. LeCun, "VICReg: Variance-invariance-covariance regularization for self-supervised learning," in *Proc. 10th Int. Conf. Learn. Representations*, 2022, pp. 1–15.
- [8] H. Sowrirajan, J. Yang, A. Y. Ng, and P. Rajpurkar, "MoCo pretraining improves representation and transferability of chest X-ray models," in *Proc. 4th Conf. Med. Imag. Deep Learn.*, 2021, vol. 143, pp. 728–744.
- [9] S. Azizi et al., "Big self-supervised models advance medical image classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 3478–3488.
- [10] Y. N. T. Vu, R. Wang, N. Balachandrar, C. Liu, A. Y. Ng, and P. Rajpurkar, "MedAug: Contrastive learning leveraging patient metadata improves representations for chest X-ray interpretation," in *Proc. 6th Mach. Learn. Healthcare Conf.*, 2021, pp. 755–769.
- [11] F. Haghighi, M. R. H. Taher, M. B. Gotway, and J. Liang, "DiRA: Discriminative, restorative, and adversarial learning for self-supervised medical image analysis," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 20824–20834.
- [12] H.-Y. Zhou, C. Lu, S. Yang, X. Han, and Y. Yu, "Preservational learning improves self-supervised medical image models by reconstructing diverse contexts," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 3499–3509.
- [13] H.-Y. Zhou, C. Lu, C. Chen, S. Yang, and Y. Yu, "A unified visual information preservation framework for self-supervised pre-training in medical image analysis," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 7, pp. 8020–8035, Jul. 2023.
- [14] K. Yan et al., "SAM: Self-supervised learning of pixel-wise anatomical embeddings in radiological images," *IEEE Trans. Med. Imag.*, vol. 41, no. 10, pp. 2658–2669, Oct. 2022.
- [15] A. Kaku, S. Upadhyay, and N. Razavian, "Intermediate layers matter in momentum contrastive self supervised learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, vol. 34, pp. 24063–24074.
- [16] K. Chaitanya, E. Erdil, N. Karani, and E. Konukoglu, "Contrastive learning of global and local features for medical image segmentation with limited annotations," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, vol. 33, pp. 12546–12558.
- [17] K. Cho et al., "CheSS: Chest X-ray pre-trained model via self-supervised contrastive learning," *J. Digit. Imag.*, vol. 36, pp. 902–910, Jan. 2023.
- [18] Y. Xie, J. Zhang, Z. Liao, Y. Xia, and C. Shen, "PGL: Prior-guided local self-supervised learning for 3D medical image segmentation," vol. abs/2011.12640, 2020, *arXiv:2011.12640*, [Online]. Available: <https://api.semanticscholar.org/CorpusID:227162415>
- [19] C. Zhang, H. Zheng, and Y. Gu, "Dive into self-supervised learning for medical image analysis: Data, models and tasks," *Med. Image Anal.*, vol. 102879, 2023, doi: [10.1016/j.media.2023.102879](https://doi.org/10.1016/j.media.2023.102879).
- [20] M. R. H. Taher, F. Haghighi, M. B. Gotway, and J. Liang, "CAiD: Context-aware instance discrimination for self-supervised learning in medical imaging," in *Proc. 5th Int. Conf. Med. Imag. Deep Learn.*, 2022, vol. 172, pp. 535–551.
- [21] V. Gorade, A. Singh, and D. Mishra, "OTCXR: Rethinking self-supervised alignment using optimal transport for chest X-ray analysis," in *Proc. 2025 IEEE/CVF Winter Conf. Appl. Comput. Vis.*, 2025, pp. 7143–7152.
- [22] V. Gorade, S. Mittal, and R. Singhal, "PACL: Patient-aware contrastive learning through metadata refinement for generalized early disease diagnosis," *Comput. Biol. Med.*, vol. 167, Dec. 2023, Art. no. 107569.
- [23] F. Sammani, B. Joukovsky, and N. Deligiannis, "Visualizing and understanding contrastive learning," *IEEE Trans. Image Process.*, vol. 33, pp. 541–555, 2024.
- [24] Y. Tian, X. Chen, and S. Ganguli, "Understanding self-supervised learning dynamics without contrastive pairs," in *Proc. 38th Int. Conf. Mach. Learn.*, 2021, vol. 139, pp. 10268–10278.
- [25] C. Shang, S. Cui, T. Li, X. Wang, Y. Li, and J. Jiang, "MATNet: Exploiting multi-modal features for radiology report generation," *IEEE Signal Process. Lett.*, vol. 29, pp. 2692–2696, 2022.
- [26] W. Zhang, B. Cai, J. Hu, Q. Qin, and K. Xie, "Visual-textual cross-modal interaction network for radiology report generation," *IEEE Signal Process. Lett.*, vol. 31, pp. 984–988, 2024.
- [27] Y. Liu, L. He, Y. Wen, L. Yang, and H. Chen, "AFiRe: Anatomy-driven self-supervised learning for fine-grained representation in radiographic images," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, pp. 18996–19005.
- [28] R. Zhai, B. Liu, A. Risteski, Z. Kolter, and P. Ravikumar, "Understanding augmentation-based self-supervised representation learning via RKHS approximation and regression," in *Proc. Int. Conf. Learn. Representations*, 2024. [Online]. Available: <https://openreview.net/forum?id=Ax2yRhCQr1>
- [29] C. J. Reed, S. Metzger, A. Srinivas, T. Darrell, and K. Keutzer, "Self-Augment: Automatic augmentation policies for self-supervised learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 2674–2683.
- [30] X. Wang and G.-J. Qi, "Contrastive learning with stronger augmentations," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 5549–5560, May 2022.
- [31] J. Zhang and K. Ma, "Rethinking the augmentation module in contrastive learning: Learning hierarchical augmentation invariance with expanded views," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 16650–16659.
- [32] S.-C. Huang, A. Pareek, M. Jensen, M. P. Lungren, S. Yeung, and A. S. Chaudhari, "Self-supervised learning for medical image classification: A systematic review and implementation guidelines," *NPJ Digit. Med.*, vol. 6, no. 1, 2023, Art. no. 74.
- [33] C. Zhang, H. Zheng, and Y. Gu, "Dive into the details of self-supervised learning for medical image analysis," *Med. Image Anal.*, vol. 89, 2023, Art. no. 102879.
- [34] M. Zheng et al., "ReSSL: Relational self-supervised learning with weak augmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, vol. 34, pp. 2543–2555.
- [35] X. Chen and K. He, "Exploring simple siamese representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 15750–15758.
- [36] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, and R. M. Summers, "ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 2097–2106.
- [37] G. Shih et al., "Augmenting the national institutes of health chest radiograph dataset with expert annotations of possible pneumonia," *Radiol. Artif. Intell.*, vol. 1, no. 1, 2019, Art. no. e180041.
- [38] Z. Anna et al., "SIIM-ACR Pneumothorax Segmentation," 2019. [Online]. Available: <https://www.kaggle.com/competitions/siim-acr-pneumothorax-segmentation/>
- [39] Y. You, I. Gitman, and B. Ginsburg, "Large batch training of convolutional networks with Layerwise Adaptive Rate Scaling," in *Proc. Int. Conf. Learn. Representations*, 2017. [Online]. Available: <https://openreview.net/forum?id=rJ4uaX2aW>
- [40] A. Singh, V. Gorade, and D. Mishra, "MLVICX: Multi-level variance-covariance exploration for chest X-ray self-supervised representation learning," *IEEE J. Biomed. Health Inform.*, vol. 28, no. 12, pp. 7480–7490, Dec. 2024.
- [41] A. Singh and D. Mishra, "CoBooM: Codebook guided bootstrapping for medical image representation learning," in *Proc. Med. Image Comput. Comput.-Assist. Intervention*, Cham, Switzerland, Oct. 2024, pp. 22–33.
- [42] T. Chen, S. Kornblith, K. Swersky, M. Norouzi, and G. E. Hinton, "Big self-supervised models are strong semi-supervised learners," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, vol. 33, pp. 22243–22255.
- [43] X. Chen, S. Xie, and K. He, "An empirical study of training self-supervised vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 9640–9649.
- [44] Q. Cai, Y. Wang, Y. Pan, T. Yao, and T. Mei, "Joint contrastive learning with infinite possibilities," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, vol. 33, pp. 12638–12648.